Supplement: Next-Day Municipal Service-Demand Forecasting Across Four Cities

Nassar Naif Alsharif

June 2026

Contents

1 Extended Notes on Data Provenance

Sources and licenses.

- **New York**: *311 Service Requests* (current dataset, scoped 2020 to present — which sets the benchmark's uniform 2020-01-01 window start, decision D14), NYC Open Data, dataset `erm2-nwe9`; NYC Open Data terms (free use without restriction).
- **Chicago**: *311 Service Requests*, Chicago Data Portal, dataset `v6vf-nfxy`; covers the current 311 system live since 2018-12-18 (the uniform study window starts 2020-01-01 per decision D14, well clear of the system transition); source-flagged duplicates dropped.
- **San Francisco**: *311 Cases*, DataSF, dataset `vw6y-z8j6`; Public Domain Dedication and License.
- **Austin**: *Austin 311 Public Data*, City of Austin Open Data Portal, dataset `xwdj-i9he`.

- **Weather**: NOAA GHCN-Daily through the AWS Open Data Program mirror `noaa-ghcn-pds` (maintained by NOAA NCEI); stations USW00094728 (Central Park), USW00094846 (O'Hare), USW00023272 (San Francisco Downtown), USW00013904 (Camp Mabry). Native units (tenths of mm/°C, tenths of m/s) converted only in preprocessing.

Acquisition queries. Each 311 source is queried through the SODA API with server-side aggregation — `$select=date_trunc_ymd(created_date), category, count(*)`, grouped and ordered, chunked by calendar year, paginated at 50,000 rows — so the rawest stored layer is already free of record-level attributes. The complete list of issued query URLs for every city is in `data/raw/311/*_manifest.json`.

Known data characteristics. Reported request volumes reflect intake-channel structure (phone/app/web), administrative practice, and reporting propensity, all of which differ across cities and over time; the Chicago information-only and aircraft-noise exclusions and the SF duplicate exclusion are documented with reasons in `configs/service_families.yml`. A missing (day, family) pair within a city's observed range denotes zero requests created, and the panel densification step materializes those zeros explicitly.

2 Service-Family Harmonization

The harmonization rule set (`configs/service_families.yml`) is an ordered, case-insensitive substring matcher: the first matching family wins, and categories matching no rule fall to *other*. Order resolves genuine ambiguities deterministically (e.g., "noise" complaints about construction are noise, not housing/building, because the noise rule precedes). The full mapping from every native category to its family, with request counts, is exported per city to `outputs/tables/family_composition_{city}.csv` by the preprocessing stage and ships with the repository; the tables below summarize the dominant native categories per family in each city.

Two properties matter for interpretation. First, the rules are *versioned*: any change to the mapping is a visible diff, and the unit test suite pins representative mappings. Second, family compositions are *not* identical across cities — e.g., the noise family is dominated by residential-noise complaints in New York but is a much smaller share elsewhere; the sanitation family includes encampment-related categories in San Francisco that have no Chicago counterpart. All cross-city statements in the main text are therefore about families as *planning buckets within each city*, and the cross-city axis compares the structure of the accuracy-to-decision mapping, never raw family volumes.

Austin

- **housing building** (1.9%): DSD - Outdoor Commercial Venue Music Complaint (7,867); Park Maintenance - Grounds Plumbing Issues (3,529); ATD - Construction Items in Right of Way (3,132)
- **noise** (4.7%): APD - Non Emergency Noise/Alarm (70,197)
- **other** (23.5%): Loose Dog (27,062); 311 CC - Other (25,843); Animal Protection - Loose Dog (24,018)
- **parks trees** (4.4%): Park Maintenance - Grounds (37,444); Tree Issue ROW (13,396); PWD - Tree Issue Right of Way (5,068)

- **public safety** (3.2%): APD - Non Emergency Collision (32,924); Sign - Traffic Sign Emergency (2,185); ZZ - AFD - Fireworks Complaint (2,049)
- **sanitation** (21.3%): ARR - Garbage (79,958); ARR - Compost (49,879); ARR - Storm Debris Collection (49,751)
- **streets transport** (38.5%): Austin Code - Request Code Officer (81,975); DSD - Request Code Officer (47,058); ATD - Parking Violation Enforcement (42,161)
- **water sewer** (2.6%): AW - Water Conservation Violation (10,628); WPD - Channels/Creek/Drainage Issues (7,643); AW - Water Waste Report (3,978)

Chicago

- **housing building** (6.9%): Building Violation (107,110); Water Lead Test Kit Request (65,635); Buildings - Plumbing Violation (40,969)
- **other** (6.9%): Stray Animal Complaint (61,065); Report an Injured Animal (50,921); Vicious Animal Complaint (37,396)
- **parks trees** (10.3%): Tree Trim Request (NO LONGER BEING ACCEPTED) (206,525); Weed Removal Request (108,205); Tree Removal Inspection (96,345)
- **public safety** (2.6%): Tree Emergency (121,023); Commercial Fire Safety Inspection Request (1,557)
- **sanitation** (38.3%): Graffiti Removal Request (556,086); Rodent Baiting/Rat Complaint (295,754); Garbage Cart Maintenance (282,964)
- **streets transport** (28.7%): Abandoned Vehicle Complaint (269,189); Pothole in Street Complaint (215,010); Street Light Out Complaint (197,946)
- **water sewer** (6.3%): Water in Basement Complaint (56,251); Sewer Cleaning Inspection Request (52,303); Check for Leak (45,813)

New York

- **housing building** (20.5%): HEAT/HOT WATER (1,439,229); UNSANITARY CONDITION (605,801); PLUMBING (343,419)
- **noise** (23.3%): Noise - Residential (2,287,530); Noise - Street/Sidewalk (1,027,860); Noise - Vehicle (374,275)
- **other** (6.2%): GENERAL (166,982); Consumer Complaint (138,005); Maintenance or Facility (121,806)
- **parks trees** (2.9%): Damaged Tree (214,620); Overgrown Tree/Branches (115,202); New Tree Request (93,700)
- **public safety** (2.7%): Non-Emergency Police Matter (190,623); Illegal Fireworks (112,922); Drug Activity (86,992)
- **sanitation** (11.5%): Request Large Bulky Item Collection (632,148); Dirty Condition (224,713); Rodent (220,299)

- **streets transport** (28.5%): Illegal Parking (2,478,000); Blocked Driveway (934,597); Street Condition (410,510)
- **water sewer** (4.4%): Water System (366,096); WATER LEAK (220,757); Sewer (172,464)

San Francisco

- **housing building** (0.6%): Residential Building Request (11,712); General Request - BUILDING INSPECTION (7,468); Residential Building (6,364)
- **noise** (1.3%): Noise Report (37,785); Noise (18,133)
- **other** (8.8%): General Request - PUBLIC WORKS (105,723); General Request (62,711); General Request - MTA (54,573)
- **parks trees** (2.8%): Tree Maintenance (80,947); Rec and Park Requests (40,539)
- **public safety** (0.0%): General Request - FIRE DEPARTMENT (490)
- **sanitation** (19.6%): Graffiti (337,617); Encampments (158,952); Graffiti Public (121,575)
- **streets transport** (65.4%): Street and Sidewalk Cleaning (1,933,456); Parking Enforcement (602,068); Abandoned Vehicle (62,042)
- **water sewer** (1.6%): Sewer Issues (47,683); Sewer (18,406); Catch Basin Maintenance (2,161)

3 Model Configurations

Hyperparameters are fixed across cities and feature sets so that no configuration receives a larger tuning budget than another; values were chosen a priori in the ranges standard for daily-count panels of this size and are not per-task optimized.

Model	Parameters	Notes
Trailing mean	7-day window	forecast-free baseline
Seasonal naive	same weekday, $t-6$	forecast-free baseline
Ridge	$\alpha = 1.0$, standardized inputs	
Random forest	300 trees, min leaf 5, 60% features	Breiman (2001)
LightGBM point	L1 objective, 600 trees, lr 0.05, 63 leaves, min child 30	Ke et al. (2017)
LightGBM quantile	as above, pinball objective per level non-crossing via cumulative max	levels .05/.25/.50/.75/.95

As the target is a non-negative count, a Poisson GLM (log link, standardized inputs) is reported as an additional count-data baseline (`tab21_poisson`); fit per city and feature set on train+validation and evaluated once on test, it beats the naive trailing-mean in 11 of 12 (city, feature set) cells but is dominated by the validation-selected gradient-boosted and tree models in all 12, so it is reported but is not a candidate in the pre-registered selection or decision grids.

Categorical handling: one-hot family indicators everywhere; one-hot city indicators in pooled (global) models only. Predictions are clipped at zero. The selected configuration per (scope, city, feature set, model) is refit on train+validation before its single test evaluation, exactly as in the protocol of the main text.

4 Allocation Optimality

The greedy allocation used by all expected-value policies is exactly optimal for the per-day problem it solves.

Proposition 1. *Let $D_1, \ldots, D_S$ be non-negative random variables, $\omega_s > 0$, $\kappa > 0$, and consider*

$$\max_{x \in \mathbb{Z}^S_{\geq 0},\, \sum_s x_s = B} \sum_{s=1}^{S} \omega_s \, \mathbb{E}[\min(D_s, \kappa x_s)] \, .$$

The greedy algorithm that assigns capacity units one at a time to the family with the largest marginal gain attains the optimum.

Proof. For fixed s, define $g_s(x) = \omega_s \mathbb{E}[\min(D_s, \kappa x)]$ on integers $x \geq 0$. For any realization $d \geq 0$ the map $c \mapsto \min(d, c)$ is concave in c; expectation preserves concavity, and restriction to the integer grid yields non-increasing increments $\Delta_s(x) = g_s(x+1) - g_s(x)$. The problem is therefore the maximization of a separable discrete-concave function over the simplex $\sum_s x_s = B$, for which the greedy/incremental algorithm is optimal: at every step the chosen increment is at least as large as any increment available later (by monotonicity of Δ_s), so an exchange argument converts any optimal solution into the greedy one without loss of objective value. □

In the implementation $\mathbb{E}[\min(D_s, c)]$ is evaluated exactly for the discrete distribution defined by linear interpolation of the inverse CDF through the five forecast quantiles on a fixed 99-point grid (`src/optimization/allocation.py`); point forecasts use a degenerate distribution, and the hindsight myopic reference uses the realized value. The unit test `test_greedy_matches_bruteforce_on_small_instance` verifies greedy-vs-brute-force equality on random instances.

The piecewise-flat marginal-value profile of degenerate forecasts (mechanism, both segments). For a degenerate distribution at point value d, $\mathbb{E}[\min(D, \kappa x)] = \min(d, \kappa x)$, so the marginal value of each additional unit equals $\kappa\omega$ while $\kappa x < d$ (the deficit segment) and 0 once $\kappa x \geq d$ (the surplus segment). Under an equal-weight objective the greedy ordering therefore carries no information in *either* segment, and allocation is decided by tie-breaking. Worked examples (computed with the repository's own `greedy_allocate`): *surplus* — budget 12, three families with point demand 30 each, $\kappa = 10$: greedy returns $[6, 3, 3]$, piling the surplus uninformatively; *deficit* — budget 4 with the same demands: greedy returns $[3, 1, 0]$ versus proportional $[2, 1, 1]$; with unequal demands 50/30/10 and budget 4, greedy returns $[4, 0, 0]$ versus proportional $[2, 1, 1]$. Distributional tails make $\mathbb{E}[\min(D, \kappa x)]$ strictly concave and restore an informative gradient in both segments.

5 Full Result Tables

All tables regenerate from `scripts/make_tables_figures.py`; CSV versions live in `outputs/tables/` with provenance hashes in `_provenance.json`.

Validation-selected headline (test MAE of the validation-chosen model per city and feature set):

city	internal	calendar	weather	calendar_weather
austin	16.529	14.843	16.530	14.844
chicago	60.374	50.945	58.665	49.295
nyc	207.022	182.378	186.728	165.417
sf	35.183	31.532	35.866	30.061

Selection detail (chosen model, validation MAE, single test evaluation):

city	feature_set	selected_model	val_mae	test_mae
austin	calendar	lgbm_point	15.389	14.843
austin	calendar_weather	lgbm_point	15.245	14.844
austin	calendar_weather_lagged_only	lgbm_point	15.711	14.865
austin	internal	random_forest	16.494	16.529
austin	weather	lgbm_point	16.462	16.530
chicago	calendar	lgbm_point	48.527	50.945
chicago	calendar_weather	lgbm_point	46.802	49.295
chicago	calendar_weather_lagged_only	lgbm_point	47.997	50.388
chicago	internal	random_forest	54.803	60.374
chicago	weather	lgbm_point	52.580	58.665
nyc	calendar	lgbm_point	146.497	182.378
nyc	calendar_weather	lgbm_point	131.660	165.417
nyc	calendar_weather_lagged_only	lgbm_point	150.323	178.626
nyc	internal	lgbm_point	165.701	207.022
nyc	weather	lgbm_point	153.963	186.728
sf	calendar	lgbm_point	30.341	31.532
sf	calendar_weather	lgbm_point	28.973	30.061
sf	calendar_weather_lagged_only	lgbm_point	30.173	31.659
sf	internal	ridge	31.435	35.183
sf	weather	ridge	31.850	35.866

Exhaustive descriptive grid (every model × feature set; never used for selection):

city	model	internal	calendar	weather	calendar_weather
austin	lgbm_point	16.719	14.843	16.530	14.844
austin	naive_trailing7	30.371	30.371	30.371	30.371
austin	random_forest	16.529	15.506	16.617	15.467
austin	ridge	24.976	25.677	24.955	25.609
austin	seasonal_naive7	21.613	21.613	21.613	21.613
chicago	lgbm_point	59.236	50.945	58.665	49.295
chicago	naive_trailing7	111.461	111.461	111.461	111.461
chicago	random_forest	60.374	54.418	59.268	53.741
chicago	ridge	71.809	70.510	72.851	70.810
chicago	seasonal_naive7	84.238	84.238	84.238	84.238
nyc	lgbm_point	207.022	182.378	186.728	165.417
nyc	naive_trailing7	305.618	305.618	305.618	305.618
nyc	random_forest	209.281	195.088	200.152	190.629
nyc	ridge	215.154	213.486	214.625	213.113
nyc	seasonal_naive7	282.439	282.439	282.439	282.439
sf	lgbm_point	35.046	31.532	33.225	30.061
sf	naive_trailing7	44.769	44.769	44.769	44.769
sf	random_forest	34.515	32.840	33.215	31.562
sf	ridge	35.183	36.562	35.866	36.691
sf	seasonal_naive7	43.786	43.786	43.786	43.786

Quantile-forecast evaluation (one unremarked pattern worth noting: the pooled quantile model's 90% test coverage, 0.745–0.872, is closer to nominal than the local model's in three of four cities, even though the local model is the decision layer's input):

scope	city	model	pinball	coverage_90	coverage_50	mae_median
local	austin	lgbm_quantile	5.162	0.812	0.413	14.790
local	chicago	lgbm_quantile	17.285	0.768	0.375	48.959
local	nyc	lgbm_quantile	58.980	0.769	0.342	167.194
local	sf	lgbm_quantile	10.189	0.808	0.344	29.286
global	austin	lgbm_quantile	4.991	0.872	0.485	15.169
global	chicago	lgbm_quantile	17.618	0.831	0.456	52.561
global	nyc	lgbm_quantile	58.862	0.818	0.408	170.492
global	sf	lgbm_quantile	9.748	0.745	0.399	29.611

Rank agreement between accuracy and simulated-decision orderings (descriptive):

city	regime	n_models	spearman_rho
austin	generous	11	0.991
austin	moderate	11	0.991
austin	scarce	11	0.982
chicago	generous	11	0.955
chicago	moderate	11	0.936
chicago	scarce	11	0.955
nyc	generous	11	0.964
nyc	moderate	11	0.964
nyc	scarce	11	0.964
sf	generous	11	0.882
sf	moderate	11	0.900
sf	scarce	11	0.900

Validation selection by MAE versus by decision loss (frozen budgets):

city	regime	selected_by_val_mae	selected_by_val_decision	test_loss_mae_choice	test_loss_decision_choice	mean_daily_diff	ci95_lo	ci95_hi	verdict
austin	scarce	global/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	16572872.000	16555517.000	-62.880	-133.118	-12.916	gain
austin	moderate	global/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	10869184.000	10857743.000	-41.453	-99.875	-0.543	gain
austin	generous	global/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	7085533.000	7076314.000	-33.402	-99.887	16.009	tie_ci_overlaps_zero
chicago	scarce	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	38101878.000	38101878.000	0.000	0.000	0.000	tie_same_choice
chicago	moderate	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	21402749.000	21402749.000	0.000	0.000	0.000	tie_same_choice
chicago	generous	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	6920807.000	6920807.000	0.000	0.000	0.000	tie_same_choice
nyc	scarce	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	154437576.000	154437576.000	0.000	0.000	0.000	tie_same_choice
nyc	moderate	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	91571785.000	91571785.000	0.000	0.000	0.000	tie_same_choice
nyc	generous	local/calendar_weather/lgbm_point	local/calendar_weather/lgbm_point	27875499.000	27875499.000	0.000	0.000	0.000	tie_same_choice
sf	scarce	global/calendar_weather/lgbm_point	global/calendar_weather/lgbm_point	30255010.000	30255010.000	0.000	0.000	0.000	tie_same_choice
sf	moderate	global/calendar_weather/lgbm_point	global/calendar_weather/lgbm_point	16972510.000	16972510.000	0.000	0.000	0.000	tie_same_choice
sf	generous	global/calendar_weather/lgbm_point	global/calendar_weather/lgbm_point	3952067.000	3952067.000	0.000	0.000	0.000	tie_same_choice

The complete decision grid (including the labeled, non-bounding hindsight myopic diagnostic) is `tab3_decision_all.csv`; block-bootstrap intervals for the pre-named contrasts are `tab8_inference.csv` (decision) and `tab9_forecast_inference.csv` (forecast), each at block lengths 14/28/56.

Dataset audit. Acquired, excluded, and retained records and native-category counts per city (from the build manifests):

City	Dataset id	Retrieved	Native cats	Acquired	Excluded	Retained
New York	erm2-nwe9	2026-06-09	271	19669371	0	19669371
Chicago	v6vf-nfxy	2026-06-09	107	11309296	6625786	4683510
San Francisco	vw6y-z8j6	2026-06-09	136	4374676	0	4374676
Austin	xwdj-i9he	2026-06-09	346	1487588	0	1487588
Total			860	36840931	6625786	30215145

Chronological split boundaries. Exact per-city train / validation / test day ranges. All feature sets share a single weather-joined evaluation panel. The NOAA GHCN-Daily layer acquired for this benchmark spans 2019-01-01 to 2025-02-06 for the four stations; because the panel requires weather features, the modeled window for every feature set runs through 2025-02-05. The 311 layer covers through 2025-12-31, so later 311 observations are retained in the densified panel but lie outside this frozen common evaluation window and are not modeled:

City	Train	Validation	Test
Austin	2020-01-28 .. 2023-08-04	2023-08-05 .. 2024-05-05	2024-05-06 .. 2025-02-05
Chicago	2020-01-28 .. 2023-08-03	2023-08-04 .. 2024-05-04	2024-05-05 .. 2025-02-04
New York	2020-01-28 .. 2023-08-03	2023-08-04 .. 2024-05-04	2024-05-05 .. 2025-02-04
San Francisco	2020-01-28 .. 2023-07-29	2023-07-30 .. 2024-05-06	2024-05-07 .. 2025-02-05

Eleven-configuration cross-scope grid. The pre-specified candidate set for the main text's Section VII selection experiment is, exactly: ten local configurations — {naive trailing-7, seasonal-naive-7, ridge, random forest, LightGBM}×internal; {random forest, LightGBM}×calendar; {ridge, random forest, LightGBM}×calendar+weather — plus the stage-censored global LightGBM on calendar+weather. No quantile model is a candidate.

Objective non-identification (full grid). The compact main-text tie-break table (main paper) reports the tie rate and the point-forecast greedy loss under five tie-break rules; the full grid of every tie-break rule and control policy across all 12 city-regime cells is `tab12_tiebreak_full.csv`, and the per-cell tie frequency is `tie_frequency.csv`.

Conformal recalibration. Empirical 90% coverage and mean interval width before and after split-conformal recalibration, and the calibrated-vs-uncalibrated full-arm decision losses:

city	coverage90_uncal	coverage90_conformal	width90_uncal	width90_conformal	delta_q05	delta_q95
austin	0.812	0.903	56.217	69.526	-3.468	9.845
chicago	0.768	0.899	172.383	207.667	-11.618	23.928
nyc	0.769	0.883	503.653	642.612	-19.792	119.167
sf	0.808	0.894	90.932	101.241	-4.215	6.691

city	regime	units	full_uncal_loss	full_conformal_loss	proportional_loss	conformal_vs_uncal_pct	conformal_vs_prop_pct
austin	scarce	8	16410288.000	16398015.000	16383102.000	-0.075	0.091
austin	moderate	11	10679297.000	10667024.000	10651410.000	-0.115	0.147
austin	generous	13	6856849.000	6844424.000	6838453.000	-0.181	0.087
chicago	scarce	29	37208413.000	37183378.000	37162683.000	-0.067	0.056
chicago	moderate	38	20087932.000	20040119.000	19965951.000	-0.238	0.371
chicago	generous	46	5561653.000	5546422.000	5451400.000	-0.274	1.743
nyc	scarce	119	146938634.000	146813066.000	146024681.000	-0.085	0.540
nyc	moderate	153	82149267.000	81984117.000	81040481.000	-0.201	1.164
nyc	generous	187	18748666.000	18583257.000	17595851.000	-0.882	5.612
sf	scarce	26	29644860.000	29642224.000	29492173.000	-0.009	0.509
sf	moderate	33	16362360.000	16359724.000	16204987.000	-0.016	0.955
sf	generous	41	3693238.000	3690775.000	3733466.000	-0.067	-1.143

Normalized pooling. Pooled-minus-local test MAE (% of local) under raw, log1p, and per-city-standardized targets:

city	pooled_log1p_pct_vs_local	pooled_per_city_z_pct_vs_local	pooled_raw_pct_vs_local
austin	1.980	2.030	4.100
chicago	5.050	4.020	6.500
nyc	2.100	1.560	1.800
sf	-0.560	-2.410	-1.170

Horizon robustness. Policy ranking at the first 30/60/90/180 and full test days is `tab15_horizon.csv`; the claim-bearing orderings (uniform worst; fixed-index point-greedy below all identified policies) are stable across horizons, while the three near-tied identified policies reshuffle within $< 1\%$.

Per-family served fraction. Served fraction by family under the uniform floor and the full-distribution arm (moderate regime), so any systematic per-family deprioritization is visible:

city	family	quantile/full_arm	uniform
austin	housing_building	0.997	1.000
austin	noise	0.997	1.000
austin	other	0.999	0.589
austin	parks_trees	0.993	1.000
austin	public_safety	0.998	1.000
austin	sanitation	0.997	0.291
austin	streets_transport	0.186	0.183
austin	water_sewer	0.000	1.000
chicago	housing_building	0.998	1.000
chicago	other	0.998	1.000
chicago	parks_trees	0.997	0.998
chicago	public_safety	0.999	1.000
chicago	sanitation	1.000	0.286
chicago	streets_transport	0.673	0.426
chicago	water_sewer	0.000	1.000
nyc	housing_building	0.999	0.455
nyc	noise	1.000	0.402
nyc	other	0.999	1.000
nyc	parks_trees	1.000	1.000
nyc	public_safety	0.999	1.000
nyc	sanitation	0.982	1.000
nyc	streets_transport	0.356	0.326
nyc	water_sewer	0.003	1.000
sf	housing_building	0.990	1.000
sf	noise	0.998	1.000
sf	other	1.000	1.000
sf	parks_trees	0.995	1.000
sf	public_safety	0.000	1.000
sf	sanitation	1.000	0.491
sf	streets_transport	0.686	0.136
sf	water_sewer	0.000	1.000

Initial-carryover sensitivity. Mean policy rank (1 = best, 5 = worst) across the 12 city-regime cells under three initial-carryover conditions ($b_0 = 0$; validation warm-up; training-average daily demand). The claim-bearing ordering is stable; full per-cell losses are in `b0_sensitivity.csv`:

policy	zero	warmup	train_avg
point_greedy_fixed	4.000	4.000	4.000
point_greedy_proportional	1.330	1.420	1.750
proportional	1.830	1.580	1.330
quantile_full	2.830	3.000	2.920
uniform	5.000	5.000	5.000

Poisson count baseline. Naive, Poisson GLM, and the validation-selected model (test MAE) per city and feature set:

city	feature_set	naive_mae	poisson_glm_mae	selected_model	selected_mae
austin	internal	30.371	31.228	random_forest	16.529
austin	calendar	30.371	26.055	lgbm_point	14.843
austin	calendar_weather	30.371	26.126	lgbm_point	14.844
chicago	internal	111.461	90.160	random_forest	60.374
chicago	calendar	111.461	74.126	lgbm_point	50.945
chicago	calendar_weather	111.461	73.525	lgbm_point	49.295
nyc	internal	305.618	263.048	lgbm_point	207.022
nyc	calendar	305.618	253.983	lgbm_point	182.378
nyc	calendar_weather	305.618	253.053	lgbm_point	165.417
sf	internal	44.769	41.869	ridge	35.183
sf	calendar	44.769	36.664	lgbm_point	31.532
sf	calendar_weather	44.769	36.336	lgbm_point	30.061

Weather missingness. Values shown are those that REMAIN missing after the quality-flag drop and the $\leq$ 7-day two-sided interpolation (they are distinct from, and additional to, the interpolation-filled values audited in the main text), per city:

City	TMAX missing	TMIN missing	PRCP missing
New York	0	0	0
Chicago	0	0	0
San Francisco	6	7	0
Austin	0	0	0

Guard suite. What each protocol guard checks and against which artifact:

Guard	Checks	Artifact
G1	Budgets from training window only; invariant to test perturbation	frozen_budgets.json
G2	Model selection uses validation MAE only; invariant to corrupted test metrics	validation_selection.csv
G3	Leave-one-city-out transfer rows carry recomputed censor dates	forecast_metrics.csv
G4	All three uncertainty arms derive from one fitted quantile model	decision_metrics.csv
G5	Structural absence (Chicago noise) never encoded as zero rows	active_families.json
G6	Allocation conserves the full budget on uneven family sets	allocation.py
G7/G8	No forbidden terminology in output tables or manuscript	tables / *.tex
G9	Every sensitivity scenario in outputs matches the config grid	decision_sensitivity.csv
G10	Dataset ids and study window agree across config and manifests	data_sources.yml / manifests
G11	Provenance SHA-256 hashes bind every table/figure to the run	_provenance.json
G12	Simulated family sets equal the active-family manifest	decision_metrics.csv
G13	Pooled training is stage-censored (fit-time proof manifest)	pooled_censoring.json
G14	Printed dataset ids are a subset of the configured ids	*.tex
G15	Table-1 caption carries the Chicago seven-family qualification	data_stats.tex
G16	Cross-scope by-MAE selection equals the recomputed argmin	decision_selection.csv

6 Negative and Conditional Results

The corrected protocol produced several results that resolved against the more optimistic hypothesis; all are retained as findings:

- **Weather adds nothing in Austin**: +0.001 test MAE on top of calendar features (block-bootstrap interval covers zero), against a 9.3% gain in New York.
- **Pooled training hurts under valid censoring**: with stage-censored pooled fits (Amendment C11), pooling raises test MAE in Austin (+4.1%), Chicago (+6.5%), and New York

(+1.8%), each 28-day block-bootstrap interval excluding zero; San Francisco is neutral. Censored zero-shot transfer degrades MAE by +5% to +149%.

- **Degenerate point forecasts non-identify the allocation objective**: under the equal-weight objective a degenerate forecast makes the marginal-value profile piecewise flat in *both* segments ($\kappa\omega$ below the point estimate plus carryover, zero above), so 92–100% of allocation steps are ties and the realized allocation is decided by the secondary tie-break rather than the forecast. Under the fixed ascending-index rule the point-forecast greedy policy underperforms proportional apportionment in all 12 city-regime settings; a proportional secondary tie-break recovers proportional behavior, and the quantile-interpolated distribution removes the non-identification and recovers much of the fixed-index loss gap (worked examples for both segments in the optimality section).

- **Quantile intervals are under-dispersed**: 90% nominal, 76.8–81.2% empirical. Split-conformal recalibration substantially improved empirical coverage (to 88–90%; calibration table above); the calibrated sensitivity did not reverse the decision-level conclusions.

- **A protocol defect was detected and corrected during review** (R1-F1): per-city split boundaries differ by 1–2 calendar days, so uncensored pooled fits contained ≈0.1% contemporaneous other-city rows. After the targeted censoring fix, the selection-experiment story changed from 7/12 disagreements with 4 apparent harms to 9/12 agreements, 2 gains, 0 harms — the earlier disagreement was substantially an artifact of the contaminated pooled model's inflated validation scores. The defect was discovered on 2026-06-11 during the documented pre-submission review-gate process (gate 1, methodologist review; finding R1-F1). Commit `e5592bb` introduced the correction (stage censoring, Amendment C11, with guard G13 and a fit-time proof manifest). All forecasts, simulations, tables, figures, and manuscript values were regenerated after the correction, and the provenance-hash guard verifies that every released manuscript artifact derives from the corrected run. The public release tag and archival DOI point only to the corrected version; pre-correction outputs are retained under `audit/pre_fix_outputs/` for audit and are not used by the paper (see `CORRECTIONS.md`).

- **The hindsight myopic reference is not a bound**: being per-day myopic under carryover it can be exceeded; it appears only in the diagnostic table below and is never used as a denominator.

7 Sensitivity Analyses

Full numbers in `outputs/tables/tab7_sensitivity.csv`.

- **Normative-priority scenario** (illustrative, not socially validated): the proportional–optimizer gap compresses relative to the equal-weights primary; the main text therefore states the proportional-dominance claim as conditional on the equal-weight objective.

- **One-family-at-a-time service yield** (×0.70 and ×1.30 for every active family, 1,125 sensitivity simulations): no policy-ordering reversal anywhere; the largest perturbation is New York, generous regime, noise yield ×1.30.

- **10% daily abandonment**: policy ordering unchanged in all four cities.

- **Budget calibration on train+validation** instead of training only: conclusions unchanged.

- **Austin without the residual *other* family** (budgets recomputed train-only over the seven retained families): identical policy ordering to the primary analysis in all three regimes; Austin's conclusions do not depend on its residual category.

- **Bootstrap block length** 14/28/56 days: 0 of 48 decision contrasts and none of the forecast contrasts change conclusion.

- **Trend diagnostic** (Newey–West lag-28 slope tests on the daily loss differentials; `tab10_trend_diagnostics`): 45 of 48 differential series and 11 of 12 loss level series are trend-dominated; only Chicago/generous is stationarity-compatible. Decision-contrast intervals are therefore read descriptively (main text, Sections VII–VIII). The diagnostic re-derives the daily series deterministically from the frozen inputs and asserts equality of every policy's total loss with the E7 artifact to 10^{-6} before computing any statistic. (San Francisco's scarce and moderate rows coincide for the three within-greedy-family contrasts: at both budgets the greedy arms are fully saturated, so each serves exactly κB per day and the paired differential depends only on the budget-invariant allocation-difference vector; the level series and the proportional contrast differ between the regimes, confirming distinct simulations.)

city	regime	contrast	n_days	mean_daily_diff	ols_slope_per_day	newey_west_t	trend_dominance_ratio	verdict
austin	scarce	full_vs_median_arm	276	-578.855	-4.558	-15.420	1.087	trending
austin	scarce	full_vs_implied_mean_arm	276	-714.163	-5.451	-12.386	1.053	trending
austin	scarce	valbest_vs_naive_greedy	276	-1810.605	-12.512	-18.676	0.954	trending
austin	scarce	greedy_vs_proportional_valbest	276	687.572	6.381	8.079	1.281	trending
austin	scarce	LEVEL_proportional_valbest	276	59359.065	393.880	34.061	NaN	trending
austin	moderate	full_vs_median_arm	276	-676.362	-5.492	-12.943	1.121	trending
austin	moderate	full_vs_implied_mean_arm	276	-875.678	-7.048	-10.288	1.111	trending
austin	moderate	valbest_vs_naive_greedy	276	-2637.562	-20.668	-56.620	1.081	trending
austin	moderate	greedy_vs_proportional_valbest	276	789.036	7.421	8.257	1.298	trending
austin	moderate	LEVEL_proportional_valbest	276	38592.065	243.881	21.090	NaN	trending
austin	generous	full_vs_median_arm	276	-872.409	-6.736	-15.431	1.066	trending
austin	generous	full_vs_implied_mean_arm	276	-1047.486	-8.630	-12.770	1.137	trending
austin	generous	valbest_vs_naive_greedy	276	-5412.554	-37.550	-36.511	0.957	trending
austin	generous	greedy_vs_proportional_valbest	276	895.217	8.550	8.611	1.318	trending
austin	generous	LEVEL_proportional_valbest	276	24777.004	143.882	12.442	NaN	trending
chicago	scarce	full_vs_median_arm	276	-3434.554	-18.827	-12.345	0.756	trending
chicago	scarce	full_vs_implied_mean_arm	276	-4633.688	-26.374	-13.059	0.785	trending
chicago	scarce	valbest_vs_naive_greedy	276	-9744.841	-60.852	-28.347	0.862	trending
chicago	scarce	greedy_vs_proportional_valbest	276	3402.880	19.789	14.643	0.803	trending
chicago	scarce	LEVEL_proportional_valbest	276	134647.402	745.261	10.606	NaN	trending
chicago	moderate	full_vs_median_arm	276	-4887.888	-34.848	-49.498	0.984	trending
chicago	moderate	full_vs_implied_mean_arm	276	-6076.783	-43.377	-43.385	0.985	trending
chicago	moderate	valbest_vs_naive_greedy	276	-14018.500	-99.213	-32.959	0.977	trending
chicago	moderate	greedy_vs_proportional_valbest	276	5205.790	38.566	43.876	1.022	trending
chicago	moderate	LEVEL_proportional_valbest	276	72340.402	295.262	4.202	NaN	trending
chicago	generous	full_vs_median_arm	276	-4911.076	-12.285	-0.886	0.345	stationary-compatible
chicago	generous	full_vs_implied_mean_arm	276	-6047.188	-16.538	-0.988	0.377	stationary-compatible
chicago	generous	valbest_vs_naive_greedy	276	-20437.344	-130.323	-9.853	0.880	trending
chicago	generous	greedy_vs_proportional_valbest	276	5324.014	13.986	0.917	0.363	stationary-compatible
chicago	generous	LEVEL_proportional_valbest	276	19751.373	-51.470	-0.976	NaN	stationary-compatible
nyc	scarce	full_vs_median_arm	276	-29346.533	-213.706	-87.992	1.005	trending
nyc	scarce	full_vs_implied_mean_arm	276	-34431.540	-246.049	-50.258	0.986	trending
nyc	scarce	valbest_vs_naive_greedy	276	-54137.612	-382.018	-55.941	0.974	trending
nyc	scarce	greedy_vs_proportional_valbest	276	30481.504	220.042	62.179	0.996	trending
nyc	scarce	LEVEL_proportional_valbest	276	529074.931	3961.562	64.340	NaN	trending
nyc	moderate	full_vs_median_arm	276	-36728.659	-286.308	-75.131	1.076	trending
nyc	moderate	full_vs_implied_mean_arm	276	-42909.221	-326.722	-91.909	1.051	trending
nyc	moderate	valbest_vs_naive_greedy	276	-68753.630	-511.885	-146.544	1.027	trending
nyc	moderate	greedy_vs_proportional_valbest	276	38156.899	297.572	72.649	1.076	trending
nyc	moderate	LEVEL_proportional_valbest	276	293624.931	2261.562	36.730	NaN	trending
nyc	generous	full_vs_median_arm	276	-35670.877	-308.964	-30.561	1.195	trending
nyc	generous	full_vs_implied_mean_arm	276	-41400.859	-350.388	-38.135	1.168	trending
nyc	generous	valbest_vs_naive_greedy	276	-71800.659	-558.941	-80.896	1.074	trending
nyc	generous	greedy_vs_proportional_valbest	276	37197.471	329.065	26.151	1.221	trending
nyc	generous	LEVEL_proportional_valbest	276	63800.714	566.092	9.329	NaN	trending
sf	scarce	full_vs_median_arm	275	-2518.247	-16.359	-20.023	0.893	trending
sf	scarce	full_vs_implied_mean_arm	275	-2658.233	-17.806	-24.147	0.921	trending
sf	scarce	valbest_vs_naive_greedy	275	-4528.535	-31.304	-42.528	0.950	trending
sf	scarce	greedy_vs_proportional_valbest	275	2772.956	17.481	16.062	0.867	trending
sf	scarce	LEVEL_proportional_valbest	275	107245.262	871.443	40.272	NaN	trending
sf	moderate	full_vs_median_arm	275	-2518.247	-16.359	-20.023	0.893	trending
sf	moderate	full_vs_implied_mean_arm	275	-2658.233	-17.806	-24.147	0.921	trending
sf	moderate	valbest_vs_naive_greedy	275	-4528.535	-31.304	-42.528	0.950	trending
sf	moderate	greedy_vs_proportional_valbest	275	2755.022	17.480	16.066	0.872	trending
sf	moderate	LEVEL_proportional_valbest	275	58963.196	521.445	24.098	NaN	trending
sf	generous	full_vs_median_arm	275	-969.404	-10.830	-9.321	1.536	trending
sf	generous	full_vs_implied_mean_arm	275	-968.091	-11.890	-7.966	1.689	trending
sf	generous	valbest_vs_naive_greedy	275	-2467.349	-25.675	-15.633	1.431	trending
sf	generous	greedy_vs_proportional_valbest	275	833.004	10.344	7.062	1.707	trending
sf	generous	LEVEL_proportional_valbest	275	13538.149	150.701	12.599	NaN	trending

Global κ granularity sweep. We rerun the complete decision evaluation at $\kappa \in \{1, 5, 10, 25, 50, 100\}$, recomputing each city–regime integer budget from the same frozen training-window formula and unchanged training means, with identical forecasts, objective, carryover, weights, regimes, seeds, and tie-break definitions (no retuning). All four cities, three regimes, and every policy are included; the fast incremental-gain greedy used here is verified equal to the published `greedy_allocate` on randomized instances, and the $\kappa = 50$ cells reproduce the committed `tiebreak_sensitivity` values. The claim-bearing ordering holds in all 72 city–regime–κ cells: uniform is worst and fixed-index point-greedy is below the identified policies (proportional, proportional tie-break, full distribution) in every cell. Magnitudes (`tab23_kappa_summary`, full grid `tab24_kappa_full`): the tie share stays near-saturated at fine granularity (99.5% mean at $\kappa = 1$, 96.6% at $\kappa = 100$) and the fixed-index penalty is *largest* at $\kappa = 1$ (e.g., New York/generous -72.2% vs. -58.3% at $\kappa = 50$), so the tie pathology is most consequential under fine, not coarse, discrete units. The proportional secondary rule stays within 0.2% of direct proportional across the sweep, and the full-distribution arm never falls below proportional in aggregate (it is not claimed to dominate it).

Fixed-index penalty (% vs. proportional; negative = worse) by κ:

City	Regime	kappa=1	kappa=5	kappa=10	kappa=25	kappa=50	kappa=100
austin	generous	-16.4	-13.3	-10.8	-7.2	-3.6	-7.7
austin	moderate	-7.8	-6.5	-5.1	-3.1	-2.0	-7.7
austin	scarce	-5.3	-4.3	-3.4	-2.3	-1.2	-7.7
chicago	generous	-47.3	-44.3	-41.7	-34.7	-27.0	-19.1
chicago	moderate	-12.2	-11.5	-10.7	-8.7	-7.2	-5.2
chicago	scarce	-5.4	-5.0	-4.6	-3.6	-2.5	-1.9
nyc	generous	-72.2	-71.3	-69.9	-66.5	-58.3	-54.6
nyc	moderate	-15.8	-15.5	-15.2	-14.4	-13.0	-11.4
nyc	scarce	-6.8	-6.7	-6.6	-6.2	-5.8	-5.1
sf	generous	-18.7	-16.4	-14.0	-10.6	-6.2	-2.3
sf	moderate	-10.4	-9.3	-8.4	-6.5	-4.7	-3.5
sf	scarce	-5.5	-5.0	-4.4	-3.6	-2.6	-1.8

Tie share (% of sequential allocation steps) by κ:

City	Regime	kappa=1	kappa=5	kappa=10	kappa=25	kappa=50	kappa=100
austin	generous	100.0	100.0	100.0	100.0	99.8	93.5
austin	moderate	100.0	100.0	100.0	100.0	99.9	93.5
austin	scarce	100.0	100.0	100.0	100.0	100.0	93.5
chicago	generous	94.8	94.8	94.7	94.5	93.8	91.6
chicago	moderate	100.0	100.0	100.0	100.0	100.0	100.0
chicago	scarce	100.0	100.0	100.0	100.0	100.0	100.0
nyc	generous	99.7	99.7	99.7	99.7	99.6	99.3
nyc	moderate	100.0	100.0	100.0	100.0	100.0	100.0
nyc	scarce	100.0	100.0	100.0	100.0	100.0	100.0
sf	generous	99.7	99.6	99.4	97.5	92.1	88.6
sf	moderate	100.0	100.0	100.0	100.0	99.8	99.3
sf	scarce	100.0	100.0	100.0	100.0	99.8	99.5

(Full per-policy grid and all magnitude columns in `tab23_kappa_summary` / `tab24_kappa_full`; the $\kappa = 100$ non-monotonicity in low-volume Austin reflects the active-family budget floor binding at coarse granularity.)

Two-sided weather-interpolation causal sensitivity. Short temperature gaps ($\leq$ 7 days) are filled by two-sided linear interpolation (`build_panel.py`), which can borrow a later observation. Under the preregistered method—drop every forecast target whose weather predictors depend on an interpolated value, from every temporal stage, and refit—we find 14 TMAX and 12 TMIN interpolated values in San Francisco and two of each in Austin. Refitting the two weather-consuming feature sets on the reduced data changes test MAE by at most 1.3% in San Francisco (and 0.0% in Austin, whose affected rows fall only in the test window), leaves the validation-selected model unchanged in both cities and both feature sets, and reverses no feature-set or allocation-policy ordering (`weather_causal_sensitivity.csv`); the weather gain is not an interpolation artifact.

8 Computational Resources

All experiments run on a single Linux container (no GPU). Measured wall-clock times for the corrected rerun: 311 acquisition 29–54 minutes per run on hosted CI runners (portal-side query latency); weather acquisition under 2 minutes; panel and feature construction under 1 minute; the forecasting stage (404 evaluations including rolling-origin folds, stage-censored pooled models, the dual-fit quantile model, and censored zero-shot transfer) approximately 13 minutes; the decision stage (324 primary simulations, 1,125 sensitivity simulations, selection experiment, and 144 block-bootstrap intervals) approximately 9 minutes; inference and artifact generation under 2 minutes. Python 3.11; package versions pinned in `reproducibility/pip-freeze.txt`; global seed 20260609; the decision layer is Monte-Carlo-free.

9 Ethics and Responsible-AI Statement

Data. All inputs are public administrative aggregates (daily counts of service requests by category) and public weather observations. No individual-level, household-level, or address-level information is acquired, stored, or processed; the acquisition layer requests server-side aggregation precisely so that record-level attributes (including free-text descriptions and locations) never enter the pipeline.

People inside the data. Each request count summarizes residents' attempts to obtain public services. Two ethical asymmetries follow. First, reported volume under-represents communities with lower propensity or ability to report (Kontokosta et al., 2017; Kontokosta and Hong, 2021); optimizing service against reported volume can therefore divert resources from quiet need toward vocal demand. This study's city-level action space cannot reallocate across neighbourhoods, which contains but does not eliminate the concern: the family-level analogue (e.g., systematically deprioritizing housing complaints relative to noise) is measured and reported per policy. Second, the priority weights are normative inputs; we publish them as configuration, sweep them, and explicitly assign their real-world selection to accountable human decision-makers.

Intended and unintended use. The system evaluated here is a research benchmark. It is not a deployable dispatch tool, has no city involvement or endorsement, and the manuscript states this in every section where simulation results appear. Misuse risk (treating simulated capacity numbers as operational guidance) is mitigated by classifying all operational parameters as sensitivity-only and reporting ranges rather than recommendations.

Human oversight. The intended human-AI division of labour — forecasts and allocation suggestions with uncertainty made visible, final allocation authority retained by planners — is described in the discussion and is a design commitment of the broader research programme, not an evaluated capability of this study.

10 Reproducibility Instructions

One-command rebuild. From the repository root:

```
pip install -r reproducibility/pip-freeze.txt   # AUTHORITATIVE frozen
                     # environment (Python 3.11.15, pandas 3.0.3);
                     # requirements.txt is a broad development file only
make all             # panel -> features -> forecasts -> simulated
                     # decisions -> inference -> artifacts
python3 scripts/run_diagnostics.py   # trend diagnostic (tab10)
python3 -m pytest tests -q   # 16 guards G1-G16; 37-test suite must pass
```

The guard suite enforces the frozen protocol: training-only budget calibration invariant to test perturbation; validation-only selection invariant to corrupted test metrics; recomputed censor dates on every transfer row plus a fit-time proof manifest for pooled-model stage censoring (Amendment C11); a single fitted model behind all uncertainty arms; structural absence never encoded as zero; exact budget conservation; forbidden-terminology scans; configuration–output agreement; source-identifier consistency; and provenance hashes binding every artifact to its generating run.

Determinism. A single global seed (`20260609`) controls every stochastic component (NumPy, scikit-learn, LightGBM); expected-value allocation uses a deterministic 99-point inverse-CDF grid rather than Monte Carlo, so decision results are bit-stable given forecasts.

Data provenance. Raw acquired layers are versioned in `data/raw/` together with JSON manifests recording: source authority, dataset id and page, license note, every query URL issued, retrieval timestamps (UTC), uncompressed and compressed SHA-256 checksums, row counts, and — for 311 sources — the portal's own dataset metadata (`rowsUpdatedAt`, column inventory) at retrieval time. Re-running `make acquire-311 acquire-weather` re-derives the same layers from the sources (counts for recent periods may drift as portals backfill; the manifests pin what this paper used).

Environment. Python 3.11; exact package versions in `reproducibility/pip-freeze.txt`; no GPU required; total pipeline runtime and per-stage costs in Section 8 of this supplement.

Acquisition environment note. The research sandbox used for this project blocks direct access to the four city portals; the 311 acquisition therefore ran through the repository's GitHub Actions workflow (`.github/workflows/acquire-311-data.yml`) on GitHub-hosted runners, executing the same committed script and pushing the raw aggregates and manifests back to the branch. The first attempt with full-window queries timed out against the NYC portal and was replaced by per-calendar-year chunking; both attempts are preserved in the repository's Actions history.

References

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30*, pages 3146–3154, 2017.

Constantine E. Kontokosta and Boyeong Hong. Bias in smart city governance: How socio-spatial disparities in 311 complaint behavior impact the fairness of data-driven decisions. *Sustainable Cities and Society*, 64:102503, 2021. doi: 10.1016/j.scs.2020.102503.

Constantine E. Kontokosta, Boyeong Hong, and Kristi Korsberg. Equity in 311 reporting: Understanding socio-spatial differentials in the propensity to complain. *arXiv preprint arXiv:1710.02452*, 2017.